

Data visualization Project

DS3302

Supervised by : Dr.Waleed Al-Shareef

Group Members:

Yazan Al-Ghamdi - 444000120

Moayad Hawsawi - 444004193

Ahmed Bannani - 444003468

Sohaib Khiero - 444002658



• Introduction

Tourism is one of the key pillars of Saudi Arabia's Vision 2030, representing an essential sector for economic diversification and national development. Understanding tourism patterns—such as visitor numbers, spending behavior, and regional distribution—provides valuable insights for planning and improving tourism services.

- This project utilizes a multi-year dataset (2015–2024) covering :
 - **Number of tourists**
 - **Tourist spending**
 - **Average stay duration**
 - **Types of tourism (domestic vs inbound)**
 - **Regional tourism statistics**

The objective of this project is to transform this dataset into meaningful visual insights using Tableau through a series of analytical and interactive visualizations. These visualizations were integrated into two dashboards to illustrate temporal, spatial, and behavioral tourism patterns across Saudi Arabia .

• Data Abstraction & Overview

The dataset consists of tourism-related attributes for Saudi provinces. Data preprocessing steps included :

- **Correcting inconsistent or incorrect variable names**
- **Handling missing values**
- **Standardizing region names for geospatial visualizations**
- **Ensuring accurate data types and formats**
- **Tourism_Type : Categorical data representing the type of tourism (domestic or international)**
- **Years : Time data representing the year**

• Attributes and Data Types

E	D	B
Attribute	Type	Description
Tourists_Number	Quantitative	Total number of tourists
Overnight_Stay	Quantitative	Number of nights tourists stay
Tourists_Spending	Quantitative	Total spending
Avg_Stay	Quantitative	Average stay duration (days)
Avg_Spending_Trip	Quantitative	Average expenditure per trip
Avg_Spending_Night	Quantitative	Average expenditure per night
Province	Categorical	Province name
Tourism_Type	Categorical	Domestic or inbound
Years	Temporal	Year of observation

• Data Relationships

Several meaningful relationships exist within the dataset :

- **Tourists_Number → Tourists_Spending**
More tourists typically generate higher total spending.
- **Avg_Stay → Avg_Spending_Trip**
Longer stays generally lead to higher total expenditures.
- **Province → Spending Levels**
Provinces differ significantly in tourism activity and spending.
- **Tourism_Type → Visitor Behavior**
Domestic and inbound tourists show distinct behavioral patterns.

These relationships were explored using the appropriate visualizations.

• Task Abstraction

Temporal Analysis :

A line chart and stacked bar chart illustrate multi-year trends in tourism numbers and spending behavior.

Comparison of Tourism Types :

Comparisons between tourism types (domestic vs inbound) reveal how each category contributes to overall tourism activity.

Spatial & Geographic Analysis :

A geospatial map and a spending heatmap highlight regional tourism variations and identify high-spending or high-traffic provinces.

Behavioral Relationship Analysis :

A scatter plot shows how the duration of stay impacts spending patterns.

Distributional Analysis :

A box plot shows spending differences across regions and highlights outliers.

• Visualization Designs and Justifications

• Spending vs Stay (Scatter Plot) :

Purpose

To examine how average stay duration correlates with average spending per trip.

Encodings & Justifications

- **Position** : Avg_Stay on X-axis, Avg_Spending_Trip on Y-axis to show correlation.
- **Color** : Gradient from gold (short stay/low spending) to green (long stay/high spending).
- **Shape**: Dots to represent individual data records.
- **Size**: Uniform to avoid misleading emphasis.

Task Support

This plot clearly shows that longer stay durations lead to higher spending—helpful for evaluating spending behavior. A line chart was tested but rejected as it conceals the variability between data points.

- **Spending per Night (Box Plot)**

Purpose

To compare nightly spending across provinces and reveal distributional differences.

Encodings

- **Position** : X-axis = provinces, Y-axis = spending per night.
- **Shape** : Box plots to show median, quartiles, and outliers.
- **Color** : Gold for higher spending, green for lower spending.

Task Support

Shows which provinces have higher nightly spending and which have more variability.

Alternative Considered

A bar chart was rejected, as it cannot show spread, distribution, or outliers.

- **Tourist Mapping (Geographic Map)**

Purpose

To visualize regional distribution of tourists across the Kingdom.

Encodings

- **Shape** : Provinces displayed geographically.
- **Color** : Gold for lower values, green for higher values.
- **Intensity** : Darker color indicates higher tourist numbers.

Task Support

Shows which provinces attract the most tourists and highlights regional disparities.

Alternative Considered

A table or bar chart lacks spatial context, making the map a superior choice.

- **Tourists Trend by Year (Line Chart)**

Purpose

To track tourism spending and tourists over several years.

Encodings

- **Position:** X-axis = years, Y-axis = spending or tourists.
- **Color:** Green vs gold to differentiate tourism types.
- **Shape:** Lines to show progression over time.

Task Support :

Reveals trend patterns, impacts of specific years (e.g., pandemic), and long-term growth.

Alternative Considered : Bars were considered but rejected because line charts better show temporal continuity.

- **Distribution of Tourism by Year (Stacked Bar Chart)**

Purpose

To compare the contribution of domestic and inbound tourism annually.

Encodings

- **Position:** Years (X-axis), tourist numbers (Y-axis).
- **Color:** Gold = domestic, Green = inbound.
- **Shape:** Stacked bars to show total and breakdown.

Task Support :

Highlights how each tourism type contributes to yearly totals.

- **Spending Heatmap (Province × Year)**

Purpose

To display spending intensity across provinces over time.

Encodings

- **Rows :** Years
- **Columns:** Provinces
- **Color:** Green for high spending, gold for low spending

- **Intensity:** Represents spending magnitude

Task Support :

Shows year-to-year spending shifts and identifies regions with strong or weak tourism spending.

• **Insights and Narrative**

The collection of visualizations reveals a comprehensive story :

Regional Insights

- Makkah consistently shows the highest spending and tourist numbers.
- Riyadh and Eastern Province follow as major tourism centers.
- Border provinces reflect lower tourism activities.

Spending Behavior

- Longer stays strongly correlate with higher spending.
- Some provinces show very high nightly spending due to luxury tourism or high-value services.

Temporal Insights

- A dip in tourism appears around 2020–2021 (pandemic impact).
- Strong recovery and growth occur in 2022–2024.
- Domestic tourism dominates early years but inbound tourism shows strong long-term growth.

Dashboard Synergy

The heatmap, line chart, bar chart, and scatter plot together build a full landscape :

- Maps show **where** tourism happens
- Trends show **when** tourism grows
- Scatter plots show **why** spending increases
- Boxplots show **how** spending varies regionally

This multi-layered view provides a 360° understanding of tourism in Saudi Arabia.



• Conclusion

This project successfully transformed a complex tourism dataset into a suite of meaningful and interactive visualizations. By analyzing temporal patterns, regional differences, and behavioral relationships, the dashboards offer powerful insights into Saudi Arabia's tourism sector .

The results can support decision-makers in identifying:

- **High-value tourism regions**
- **Spending trends**
- **Growth opportunities**
- **Areas requiring investment**

Tableau proved to be a highly effective tool for performing visual analytics and building interactive dashboards that align with real-world analytical needs.

- **Recommendations**

- Improve tourism infrastructure in lower-performing provinces.
- Leverage inbound tourism potential revealed by data trends.
- Use predictive modeling to forecast spending and visitor numbers.
- Combine this dataset with hotel occupancy, flight data, and event calendars for deeper analysis.

- **References**

- Kaggle Tourism Dataset (2024) -
<https://www.kaggle.com/datasets/toobaik/saudi-arabia-tourism-dataset-20152024>
- Saudi Tourism Authority Reports
- Tableau Documentation
- Ministry of Tourism – Statistical Releases

Thank You !!

- Task Distribution:

- Yazan Al-Ghamdi (Leader):

- Team Task distribution
- Built Dashboard (Tourism Statistics Over the Years)
- created charts (Heatmap – Scatter plot)

- Ahmad Bannani:

- created charts (Geographic map – Bar Chart)
- Built Dashboard (Durational and Regional Statistics)
- Interactive elements of the dashboard

- Sohaib Khir:

- Write Report content
- Report Formatting
- Data Selection

- Moayad Hawsawi:

- created charts (Line chart– Box Plot)
- Report Insights and Narrative
- Report Headlines Formatting